

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: CSA0203

COURSE NAME: C Programming

COURSE OUTCOME COVERED

CO1: Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks. (BL2, BL3)

Assessment Tool Weightage: 10%

QUESTIONS:

Total Marks:100

Annexure A

MCQ

Code of Conduct:

I R. Jaideep (192525001) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Jaideep

ASSESSMENT TOOL 1

CO1: Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks. (BL2, BL3)

Q1. A programmer wants to display the octal equivalent of a decimal number stored as 25 using printf. Which output will the program produce?

```
int n = 25;  
printf("%o", n);
```

Output?

- ☒ A) 31
- ☐ B) 25
- ☐ C) 32
- ☐ D) Error

Q2. A developer prints an integer value 26 using the %x format specifier. Considering lowercase hexadecimal representation in C, what will be the output?

```
int n = 26;  
printf("%x", n);
```

Output?

- ☐ A) 1A
- ☒ B) 1a
- ☐ C) 26
- ☐ D) Error

Q3. Nerd initializes a variable using 017 and prints it using %d. Since the value is stored in a different number system, what will be the resulting output?

```
int n = 017;  
printf("%d", n);
```

Output?

- ☐ A) 17
- ☒ B) 15
- ☐ C) 16
- ☐ D) Error

Q4. A variable is assigned a hexadecimal constant 0xA5 and printed using %d. After decimal conversion, what value will be displayed?

```
int n = 0xA5;  
printf("%d", n);
```

Output?

- ☐ A) 155
- ☒ B) 165
- ☐ C) 150
- ☐ D) Error

Q5. Geek uses 010 as the value of an integer and prints it using %x. After converting the value appropriately, what will be the final output?

```
int n = 010;  
printf("%x", n);
```

Output?

- ☒ A) 8
- ☐ B) A
- ☐ C) 10
- ☐ D) Error

Q6. A technician adds a hexadecimal value 0x10 and an octal value 010, then prints the result using %d. What will be the output?

```
printf("%d", 0x10 + 010);
```

Output?

- ☒ A) 24
- ☐ B) 18
- ☐ C) 20
- ☐ D) 16

Q7. A variable with value 20 is printed using both %o and %x format specifiers in the same statement. What will be the output?

```
int n = 20;  
printf("%o %x", n, n);
```

Output?

- ☒ A) 24 14
- ☐ B) 20 20
- ☐ C) 24 20
- ☐ D) Error

Q8. A macro #define A 10+5 is used in an expression A*2 without parentheses. What will be the output?

```
#define A 10+5  
printf("%d", A*2);
```

Output?

- ☐ A) 30
- ☒ B) 25
- ☐ C) 40
- ☐ D) Error

Q9. A macro #define X 10 is defined, and later a variable with the same name int X = 5; is declared. What will happen when printing X?

```
#define X 10  
int X = 5;  
printf("%d", X);
```

Output?

- ☐ A) 10
- ☐ B) 5
- ☒ C) Error
- ☐ D) Garbage

Q10. A constant variable is declared as "const int x = 10" and then reassigned using "x = x + 5". What will be the result?

```
const int x = 10;
```

```
x = x + 5;
```

Result?

A) 15

B) 10

☒ C) Error

D) Garbage

Q11. A student divides an integer (5) by a float (2) and prints the result using %f. What will be the output?

```
int a = 5;
```

```
float b = 2;
```

```
printf("%f", a/b);
```

Output?

A) 2.000000

☒ B) 2.500000

C) 2

D) Error

Q12. A float variable is initialized as float x = 5/2 and then printed using %f. What will be the output?

```
float x = 5/2;
```

```
printf("%f", x);
```

Output?

A) 2.500000

☒ B) 2.000000

C) 5

D) Error

Q13. A character variable is assigned the value 100 and printed using %c. What will be displayed?

```
char c = 100;
```

```
printf("%c", c);
```

Output?

A) 100

B) d

C) Error

☒ D) Garbage

Q14. A macro, A, is defined and then undefined using #undef. What will happen when attempting to print A?

```
#define A 5
```

```
#undef A
```

```
printf("%d", A);
```

Output?

A) 5

B) 0

- ☒ C) Error
- ☐ D) Garbage

Q15. A variable is declared but not initialized, and then printed. What will be the output?

```
int a;  
printf("%d", a);
```

Output?

- ☐ A) 0
- ☐ B) Error
- ☒ C) Garbage value
- ☐ D) 1

Q16. A character, 'A', is stored in a variable, and 1 is added to it before printing the result as an integer. What will be the output?

```
char a = 'A';  
int b = a + 1;  
printf("%d", b);
```

Output?

- ☒ A) 66
- ☐ B) 65
- ☐ C) A
- ☐ D) Error

Q17. A macro `#define B 2+3` is used in the expression `B + 4`. What will be the output?

```
#define B 2+3  
printf("%d", B+4);
```

Output?

- ☒ A) 9
- ☐ B) 10
- ☐ C) 11
- ☐ D) Error

Q18. A macro `#define X 3+2` is used in the expression `X*X` without parentheses. What will be the output?

```
#define X 3+2  
printf("%d", X*X);
```

Output?

- ☐ A) 25
- ☒ B) 11
- ☐ C) 19
- ☐ D) Error

Q19. An integer is divided by a float and printed using `%f`. What will be the output of `10/3`?

```
int a = 10;  
float b = 3;  
printf("%f", a/b);
```

What will be printed?

- ☒ A) 3.333333
- ☐ B) 3.000000

- C) 3
D) Error

Q20. A character 'A' is printed using %d. What will be the output?

```
char ch = 'A';  
printf("%d", ch);
```

Output?

- A) A
B) 65
C) Error
D) Garbage

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions